Project: Worked examples
Title: Bolt group (elastic method)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the shear and tension in every bolt of a planar group under eccentric in-plane forces, torque, axial load and out-of-plane moments, by the elastic method. It also finds the governing bolt and an optional lever-model prying addition.

WHAT IT CALCULATES	RELATION
Centroid of the bolt group, each bolt weighted by its area	$x_c = \Sigma A x / \Sigma A$ (y_c likewise)
Torsion about the centroid from the eccentric load	$T = F_y (e_x - x_c) - F_x (e_y - y_c) + M_z$
Polar moment of the bolt areas about the centroid	$J = \Sigma A r^2$
Shear on each bolt, its direct share plus the torsion (vector sum)	$v = A (F/\Sigma A) + T \cdot A \cdot r / J$
Tension on each bolt, its direct share plus the moments, with the product of inertia so that an unsymmetric pattern is right	$t = A N/\Sigma A + A \cdot [(M_x I_y - M_y I_{xy}) y' + (M_y I_x - M_x I_{xy}) x'] / (I_x I_y - I_{xy}^2)$, where $I_x = \Sigma A y'^2$, $I_y = \Sigma A x'^2$, $I_{xy} = \Sigma A x'y'$ (this reduces to $M_x A y' / \Sigma A y'^2 + M_y A x' / \Sigma A x'^2$ only for a pattern symmetric about a centroidal axis)
Tension with prying, when the distances a and b are given	$t_p = t (1 + b/a)$
Combined shear and tension, circular interaction (elastic method, Shigley-style)	$(v/F_{v,Rd})^2 + (t_p/F_{t,Rd})^2 \le 1$
Combined shear and tension, EN 1993-1-8 linear form (each is also checked alone; the highest governs)	$v/F_{v,Rd} + t_p/(1.4 F_{t,Rd}) \le 1$

Basis: Design resistance of individual fasteners; elastic distribution in a bolt group[1](cl. 3.6, 3.8); Eccentrically loaded bolt groups — elastic method[2](§7, 15th ed.).

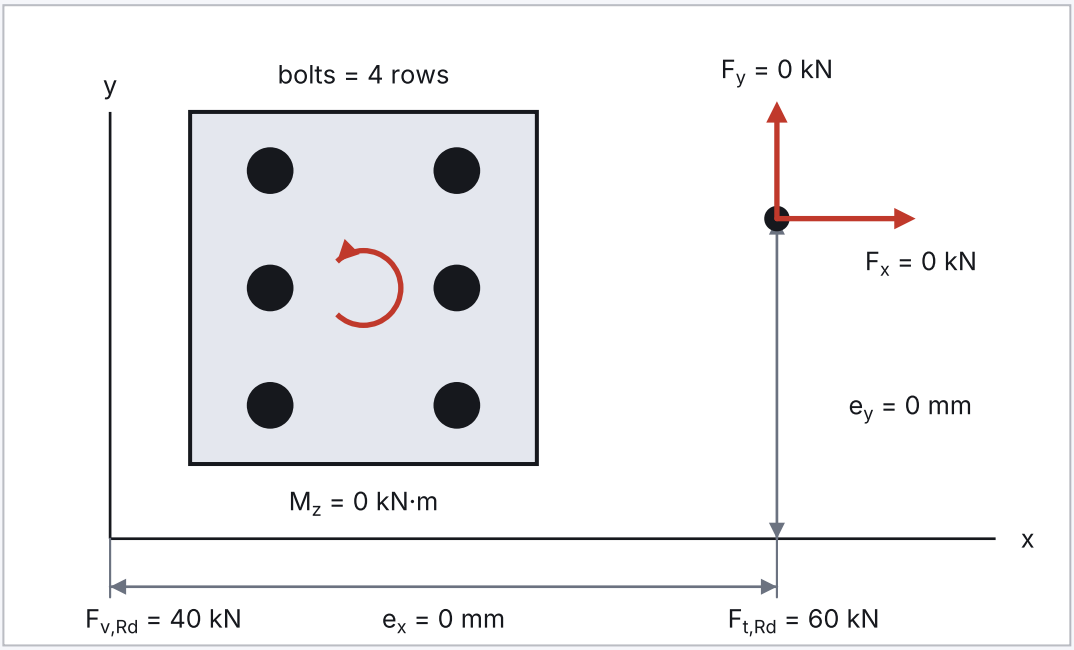


Figure 1 — A bolt group (six drawn; the bolts as listed) under an in-plane load at (e_x, e_y), x and y from one origin, + as drawn. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
bolts	Bolt positions and areas	4 row(s)		drawing: DRG-BRK-09 rev B	any
F_x	In-plane force, x	0	kN	spec: LOAD-CASE-1	-100000 … 100000 kN
F_y	In-plane force, y	0	kN	spec: LOAD-CASE-1	-100000 … 100000 kN
e_x	Point of application, x	0	mm	drawing: DRG-BRK-09 rev B	-10000 … 10000 mm
e_y	Point of application, y	0	mm	drawing: DRG-BRK-09 rev B	-10000 … 10000 mm
M_z	Additional in-plane torque about the centroid	0	kN·m	spec: LOAD-CASE-1	-100000 … 100000 kN·m
N	Axial (out-of-plane) load, tension positive	0	kN	spec: LOAD-CASE-1	-100000 … 100000 kN
M_x	Out-of-plane moment about x (tension on +y side)	20	kN·m	spec: LOAD-CASE-1	-100000 … 100000 kN·m
M_y	Out-of-plane moment about y (tension on +x side)	0	kN·m	spec: LOAD-CASE-1	-100000 … 100000 kN·m
$F_{v,Rd}$	Shear resistance per bolt	40	kN	spec: M12 8.8 shear resistance, EN 1993-1-8 Table 3.4 per project calc	0.1 … 10000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$F_{t,Rd}$	Tension resistance per bolt	60	kN	spec: M12 8.8 tension resistance per project calc	0.1 … 10000 kN

TABLE ROW	VALUES
bolts[1]	x = 0 mm; y = -100 mm; A = 84.3 mm²
bolts[2]	x = 100 mm; y = -100 mm; A = 84.3 mm²
bolts[3]	x = 0 mm; y = 100 mm; A = 84.3 mm²
bolts[4]	x = 100 mm; y = 100 mm; A = 84.3 mm²

3. Assumptions

A1. The connected parts are rigid, and bolt forces vary linearly with distance from the centroid. (The elastic method is conservative compared with plastic and instantaneous-centre methods.)
A2. Resistances are per bolt as entered, with their source (a design-code capacity or a manufacturer value). (No code capacity is computed here.)
A3. Prying uses a simple lever: the bolt tension rises by the factor (1 + b/a). (An optional, conservative first estimate.)

4. Working

[11] n_b = number of bolts
= 4
= 4
Bolts in the group
[12] $x_c = \Sigma A x / \Sigma A$
= 84.3×0 + 84.3×100 + 84.3×0 + 84.3×100 / 337.2
= 50 mm
Centroid x
[13] $y_c = \Sigma A y / \Sigma A$
= 84.3×-100 + 84.3×-100 + 84.3×100 + 84.3×100 / 337.2
= 0 mm
Centroid y
[14] $T = F_y (e_x - x_c) - F_x (e_y - y_c) + M_z$
= 0 × -50 − 0 × 0 + 0
= 0 kN·m
In-plane torque about the centroid
[15] $J = \Sigma A r^2$
= about the centroid
= 4215000 mm⁴
Polar second moment of the bolt areas
[16] b1 = bolt 1 at (0, -100) mm
= v = (0, 0) kN, t = 0 kN
= 0
Interaction utilisation $\sqrt{((v/F_{v,Rd})^2 + (t/F_{t,Rd})^2)} = 0$
[17] b2 = bolt 2 at (100, -100) mm
= v = (0, 0) kN, t = 0 kN
= 0
Interaction utilisation $\sqrt{((v/F_{v,Rd})^2 + (t/F_{t,Rd})^2)} = 0$
[18] b3 = bolt 3 at (0, 100) mm
= v = (0, 0) kN, t = 50 kN
= 0.83333
Interaction utilisation $\sqrt{((v/F_{v,Rd})^2 + (t/F_{t,Rd})^2)} = 0.833$

[19] b4 = bolt 4 at (100, 100) mm
= v = (0, 0) kN, t = 50 kN
= 0.83333
Interaction utilisation $\sqrt{((v/F_{v,Rd})^2 + (t/F_{t,Rd})^2)}$ = 0.833
[20] $F_{v,Ed,max}$ = max over the bolts of v
= bolt 1
= 0 kN
Maximum shear on any bolt
[21] $F_{t,Ed,max}$ = max over the bolts of t (with prying)
= bolt 3
= 50 kN
Maximum tension on any bolt
[22] $bolt_v$ = bolt with the largest shear
= bolt 1
= "bolt 1"
The bolt carrying the largest shear
[23] $bolt_t$ = bolt with the largest tension
= bolt 3
= "bolt 3"
The bolt carrying the largest tension
[24] $u_{max} = \sqrt{((v/F_{v,Rd})^2 + (t/F_{t,Rd})^2)}$
$= \sqrt{((0/40)^2 + (50/60)^2)}$
= 0.83333
Governing interaction utilisation
[25] governing = governing bolt
= bolt 3
= "bolt 3"
The bolt with the highest utilisation
[26] $u_v = F_{v,Ed,max} / F_{v,Rd}$
= 0 kN / 40 kN
= 0
Shear utilisation of the most sheared bolt
[27] u_{lim} = 1
= 1
Limit of an interaction utilisation
[28] $u_t = F_{t,Ed,max} / F_{t,Rd}$
= 50 kN / 60 kN
= 0.83333
Tension utilisation of the most tensioned bolt
[29] $governing_{lin}$ = bolt with the highest linear utilisation
= bolt 3
= "bolt 3"
The bolt that governs the linear shear–tension check
[30] u_{lin} = max over the bolts of $v/F_{v,Rd} + t/(1.4\ F_{t,Rd})$
= bolt 3: $0/40 + 50/(1.4 \times 60)$
= 0.59524
Linear shear–tension interaction, the highest over all bolts

5. Results

SYMBOL	VALUE	DESCRIPTION
$F_{v,Ed,max}$	0 kN	Maximum shear on any bolt
$F_{t,Ed,max}$	50 kN	Maximum tension on any bolt
T	0 kN·m	Torque about the centroid

SYMBOL	VALUE	DESCRIPTION
u_{max}	0.83	Governing interaction utilisation (governing)
J	4215000 mm⁴	Polar second moment of bolt areas
$bolt_v$	"bolt 1"	Bolt with the largest shear
$bolt_t$	"bolt 3"	Bolt with the largest tension
governing	"bolt 3"	Governing bolt
$governing_{lin}$	"bolt 3"	Bolt governing the linear interaction
u_{lin}	0.60	Linear interaction utilisation (highest over all bolts)
u_v	0.00	Shear utilisation of the most sheared bolt
u_t	0.83	Tension utilisation of the most tensioned bolt

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Circular shear–tension interaction against its limit (governing bolt)	0.83333	1	0.83	
PASS	Largest bolt tension against tension resistance	50 kN	60 kN	0.83	
PASS	Linear shear–tension interaction against its limit (every bolt)	0.59524	1	0.60	[1](Table 3.4)
PASS	Largest bolt shear against shear resistance	0 kN	40 kN	0.00	

7. Applicability

None.

8. Verification

Template mech.bolt-group@1.5.0 Content hash 6612491a876c3ed5 Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.
2. American Institute of Steel Construction. Steel Construction Manual. 15th ed. AISC; 2017.